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From jargon to clarity: Improving the readability of foot and ankle radiology reports with an artificial intelligence large language model☆

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ABSTRACT

Background: The purpose of this study was to evaluate the efficacy of an Artificial Intelligence Large Language Model (AI-LLM) at improving the readability foot and ankle orthopedic radiology reports.
Methods: The radiology reports from 100 foot or ankle X-Rays, 100 computed tomography (CT) scans and 100 magnetic resonance imaging (MRI) scans were randomly sampled from the institution's database. The following prompt command was inserted into the AI-LLM: "Explain this radiology report to a patient in layman's terms in the second person: [Report Text]". The mean report length, Flesch reading ease score (FRES) and Flesch-Kincaid reading level (FKRL) were evaluated for both the original radiology report and the AI-LLM generated report. The accuracy of the information contained within the AI-LLM report was assessed via a 5-point Likert scale. Additionally, any "hallucinations" generated by the AI-LLM report were recorded.
Results: There was a statistically significant improvement in mean FRES scores in the AI-LLM generated X-Ray report (33.8 ± 6.8 to 72.7 ± 5.4), CT report (27.8 ± 4.6 to 67.5 ± 4.9) and MRI report (20.3 ± 7.2 to 66.9 ± 3.9), all $p < 0.001$. There was also a statistically significant improvement in mean FKRL scores in the AI-LLM generated X-Ray report (12.2 ± 1.1 to 8.5 ± 0.4), CT report (15.4 ± 2.0 to 8.4 ± 0.6) and MRI report (14.1 ± 1.6 to 8.5 ± 0.5), all $p < 0.001$. Superior FRES scores were observed in the AI-LLM generated X-Ray report compared to the AI-LLM generated CT report and MRI report, $p < 0.001$. The mean Likert score for the AI-LLM generated X-Ray report, CT report and MRI report was 4.0 ± 0.3, 3.9 ± 0.4, and 3.9 ± 0.4, respectively. The rate of hallucinations in the AI-LLM generated X-Ray report, CT report and MRI report was 4%, 7% and 6%, respectively.
Conclusion: AI-LLM was an efficacious tool for improving the readability of foot and ankle radiological reports across multiple imaging modalities. Superior FRES scores together with superior Likert scores were observed in the X-Ray AI-LLM reports compared to the CT and MRI AI-LLM reports. This study demonstrates the potential use of AI-LLMs as a new patient-centric approach for enhancing patient understanding of their foot and ankle radiology reports. Jel Classifications: IV

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1. Introduction

Radiology is essential in contemporary healthcare, offering vital insights into a patient's health that may not be discernible through mere physical examination. It is estimated that over 4.2 billion radiological examinations are performed globally per year, with each examination accompanied by a radiology report most commonly generated by a radiologist [1]. Each report consists of a detailed description of the various imaging findings, technical specifications of the diagnostic tool, measurements and, occasionally, clinical recommendations. Radiology reports are complex communication tools primarily intended for physician-to-physician communication, comprised of technical medical jargon and nongrammatical fragments [2]. However, as each report is available for review by each individual patient, radiology reports have become a source of anxiety and confusion for the general patient population [3].

As a result, there has been a drive by institutions in recent years to develop patient-centric annotated radiology reports with the use of simplified terms and infographics [2]. However, these strategies have led to an increase in radiologist reporter burden and have

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produced only modest success rates. The advent of publicly available artificial intelligence large language models (AI-LLM) offers a potential solution for both the physician and the patient [4]. Leveraging natural language analysis and machine learning algorithms, the platform has the potential to identify key phrases and concepts in the report, amalgamating them into a clear and easily understandable format. However, it is currently uncertain if AI-LLMs are capable of sufficiently improving the readability of foot and ankle orthopedic radiology reports without compromising the medical accuracy of the report.

The purpose of this study was to evaluate the efficacy of an A-LLM at improving the readability foot and ankle orthopedic radiology reports. The accuracy of the AI-LLM generated report was also assessed.

2. Material and methods

2.1. Study design and data collection

After receiving Institutional Review Board approval (i23–01245), a retrospective review of all patients treated for foot and ankle pathologies by a senior board certified foot and ankle orthopedic surgeon was performed. A random selection of the reports of 100 diagnostic X-rays of the foot or ankle, 100 diagnostic computed tomography (CT) scans of the foot or ankle and 100 diagnostic magnetic resonance imaging (MRI) of the foot or ankle were identified and extracted. Each report was deidentified and subsequently inputted into the institution's AI-LLM GPT-3.5.

Two "readability" indices were utilized including the Flesch reading ease score (FRES) [5], and the Flesch-Kincaid reading level (FKRL) [6]. The FRES score is a readability metric based on 2 factors: sentence length and the number of syllables per word. The FRES score is calculated with the following formula: FRES= 206.835 − (1.015 ×average words per sentence)− (84.6 ×average syllables per word). The FRES score ranges from 0 to 100, with higher scores indicating easier readability. The FKRL score is a readability metric that measures the approximate grade level required to understand a piece of written text. The FKRL is based on 2 factors: average sentence length and average number of syllables per word. Its formula is as follows: FKRL= 0.39 ×average words per sentence+ 11.8 ×average syllables per word− 15.59. The value corresponds to the U.S grade level required to adequately understand a passage of text. For example, a FKRL of 8 indicates that the text can be understood by an individual with a Grade 8 level of education. Report length for both the raw radiology report and the corresponding AI-LLM output was calculated.

A 5 point Likert scale was created to assess the accuracy of the AI-LLM output report: 1 – "completely inaccurate"; 2 – "slightly inaccurate"; 3 – "neutral"; 4 – "accurate"; 5 – "very accurate". Each AI-LLM output report was assessed by 2 senior board certified foot and ankle orthopedic surgeons and a score was assigned to each report. Additionally, any incorrect or fabricated information (so-called "hallucinations") were identified and recorded. Hallucinations were defined as text that is contextually or factually incorrect, misleading, or unrelated to the input provided.

2.2. Data Analysis

All statistical analyses were performed using RStudio (software version 4.3.1). Mean FRES, FKRL and report length along with their respective standard deviation was recorded for each original report and its corresponding AI-LLM. A paired-2 sided test was performed to assess for significance between each individual original radiology report and its corresponding AI-LLM output report with regards to FRES, FKRL scores and report length. A one-way analysis of variance (ANOVA) and Tukey's Honestly's Statistical Difference post hoc test was performed to assess if there was any statistically significant difference in mean report length, mean FRES scores and mean FKRL scores between the original report type (X-Rays, CTs and MRIs) and the AI-LLM generated report. Fisher's exact test was calculated to assess for statistically significant difference in the hallucination rate between the 3 radiology report types. A p value of < 0.05 was considered statistically significant.

3. Results

A sample of the original report and the AI-LLM generated report for all report types is listed in Table 1. Mean FRES scores, mean FKRL scores and mean report length for the original reports are listed in Table 2.

3.1. X-Ray analysis

The mean FRES score for the original X-Ray report was 33.8 ± 6.8 which increased to a mean FRES score of 72.7 ± 5.4 following input into the AI-LLM ($p < 0.0001$). The mean FKRL score for the original X-Ray report was 12.2 ± 1.1 which decreased to a mean FKRL score of 8.5 ± 0.4 following input into the AI-LLM ($p < 0.0001$). The mean report length of the original X-Ray report was 37.6 ± 11.1 words which increased to a mean report length of 179.5 ± 33.8 following input into the AI-LLM ($p < 0.0001$).

3.2. CT analysis

The mean FRES score for the original CT report was 27.8 ± 4.6 which increased to a mean FRES score of 67.5 ± 4.9 following input into the AI-LLM ($p < 0.0001$). The mean FKRL score for the original CT report was 15.4 ± 2.0 which decreased to a mean FKRL score of 8.4 ± 0.6 following input into the AI-LLM ($p < 0.0001$). The mean report length of the original CT report was 171.4 ± 23.5 words which increased to a mean report length of 292.2 ± 16.9 following input into the AI-LLM ($p < 0.0001$).

3.3. MRI analysis

The mean FRES score for the original MRI report was 20.3 ± 7.2 which increased to a mean FRES score of 66.9 ± 3.9 following input into the AI-LLM ($p < 0.0001$). The mean FKRL score for the original MRI report was 14.1 ± 1.6 which decreased to a mean FKRL score of 8.5 ± 0.5 following input into the AI-LLM ($p < 0.0001$). The mean report length of the original MRI report was 191.6 ± 18.6 words which increased to a mean report length of 293.5 ± 13.5 following input into the AI-LLM ($p < 0.0001$).

3.4. Analysis of variance between report types

With regards to the original radiology report, one-way ANOVA demonstrated statistically significant differences in FRES scores between all 3 report types ($F = 107.5$, $p < 0.001$) (Fig. 1). There was a statistically significant higher mean FRES scores in the X-Ray reports compared to the CT reports (95% CI: [3.7, 8.0]; $p < 0.001$), in the X-Ray reports compared to MRI reports (95% CI: [11.3, 15.6]; $p < 0.001$) and CT reports compared to MRI reports (95% CI: [5.4, 9.8] $p < 0.001$). One-way ANOVA demonstrated statistically significant differences in mean FKRL scores between all report types ($F = 98.3$, $p < 0.001$) (Fig. 2). There was a statistically significant higher mean FKRL scores in the CT reports compared to the X-Ray reports (95% CI: [2.7, 3.8]; $p < 0.001$), in the CT reports compared to the MRI reports (95% CI: [1.4, 2.5]; $p < 0.001$) and MRI reports compared to the X-Ray reports (95% CI: [0.8, 1.9] $p < 0.001$). One-way ANOVA demonstrated statistically significant differences in mean report length between all report types ($F = 2054.0$, $p < 0.001$) (Fig. 3). There was a

Table 1
Comparison between original radiology report and Artificial Intelligence Large Language Model generated report.

	X-Ray foot	CT ankle	MRI ankle
Original Report	FINDINGS: Redemonstrated triple arthrodesis with screws transfixing the fused talonavicular, subtalar and calcaneocuboid articulations. Redemonstrated mature confluent osseous bridging across the subtalar and talonavicular joints. Unchanged persistent lucency at the calcaneocuboid joint with subtle lucency around the fixation screw. Moderate polyarticular osteoarthritic changes involving the naviculocuneiform, tarsometatarsal and first metatarsophalangeal-hallux sesamoid articulations, unchanged. Redemonstrated confluent ankylosis of the first interphalangeal joint. Severe talocrural osteoarthritis with tibiotalar bone-on-bone apposition, subchondral sclerosis, articular cortical irregularity of the tibial plafond and talar dome, with marginal osteophytosis. Mild soft tissue swelling about the ankle and dorsal midfoot.	FINDINGS: The examination again demonstrates that the patient is status post placement of an osteochondral graft along the medial aspect of the talar dome. There is good incorporation of the graft with the adjacent native bone. The articular surface of the graft is depressed with respect to the articular surface of the native bone by approximately 1 mm, somewhat more prominent along the lateral margin of the graft-native bone interface. There has been prior osteotomy of the base of the medial malleolus fixated with 2 partially threaded, cancellous screws. No hardware-related complication is identified. The osteotomy margin appears healed. No acute fractures identified. The joint spaces are maintained without arthritis. The bones appear somewhat demineralized, likely related to disuse. CT is not optimized for evaluation of the ligamentous or tendinous structures of the ankle, however no gross abnormalities involving these structures are seen. The subcutaneous tissues are unremarkable. IMPRESSION: 1. Status post placement of an osteochondral graft along the medial aspect of the talar dome with good incorporation of the graft into the adjacent native bone. The articular surface of the graft is depressed with respect to the articular surface of the native bone by approximately 1 mm, somewhat more prominent along the lateral margin of the graft-native bone interface. 2. Status post osteotomy the medial malleolus without evidence of complication. 3. Disuse osteopenia.	FINDINGS: Ligaments: Anterior and posterior tibiofibular ligaments are intact. Scar remodeling and attenuated anterior talofibular ligament. Remodeled calcaneofibular and posterior talofibular ligament. Remodeled deep deltoid ligament fibers. Tendons: Anterior compartment tendons are intact. Previous repair and intact tenodesis of the posterior tibial and flexor digitorum longus tendons. Intact flexor hallucis longus tendon. Intact peroneal tendons. Bones/joints: Healed calcaneal osteotomy. Postsurgical osseous remodeling at the anterior distal tibia and medial navicular. Osteochondral defect at the medial talar dome measuring 1.1 cm in length and 0.6 cm in depth with evidence of prior repair/bone graft filling. There is overlying full-thickness chondral loss with an adjacent small subchondral cyst and mild subchondral edema. Small tibiotalar osteophytes. There is bone marrow edema of the posterior talus, and about the sinus Tarsi, where there is a small intraosseous cyst at the calcaneus. There is also chondral thinning at the posterior facet of the subtalar joint with areas of osseous remodeling. Multifocal midfoot osteoarthritis including at the talonavicular and calcaneocuboid articulations. Plantar Fascia: No tear. Nerves: No mass in the tarsal tunnel. Other: Susceptibility artifact along the posteromedial hindfoot and along the plantar aspect of the medial cuneiform from tenodesis screws and metallic staple, respectively. IMPRESSION: Osteochondral defect at the medial talar dome with evidence of prior repair/bone graft filling. Overlying full-thickness chondral loss, small adjacent subchondral cyst, and mild underlying subchondral edema. Intact multiple tenodeses, including the posterior tibial tendon. Healed calcaneal osteotomy. Mild tibiotalar osteoarthritis. Multifocal midfoot osteoarthritis.
AI-LLM generated report	FINDINGS: This report is about a follow-up examination of your foot and ankle after you've had a procedure known as a triple arthrodesis. This is a surgery where three of your foot joints were fused together using screws. These joints are the talonavicular, subtalar, and calcaneocuboid joints. The report shows that these joints are still fused together, and there's new bone growth that's connecting the subtalar and talonavicular joints, which is a good sign. However, there's a small area (lucency) at the calcaneocuboid joint, where the bone isn't as dense as it should be. There's also a subtle lucency around one of the screws, which means the bone around the screw isn't as dense as it should be. You have moderate arthritis in several other joints in your foot, including the naviculocuneiform, tarsometatarsal, and the joint between your big toe and the small bones underneath it (first metatarsophalangeal-hallux sesamoid articulations). This hasn't changed since your last examination. The joint between your big toe and the rest of your	FINDINGS: This report is about a follow-up examination of your ankle after you've had two procedures done. Firstly, you had a piece of bone and cartilage (an osteochondral graft) placed on the inside part of your ankle bone (the talar dome). The graft has settled well and is now a part of your original bone. However, the surface of the graft is slightly lower (by about 1 millimeter) than the surface of your original bone, and this is more noticeable on the side of the graft that's closer to the outside of your ankle. Secondly, you had a surgical cut (osteotomy) made at the base of the bump on the inside of your ankle (the medial malleolus). This was held in place with two screws. There are no complications from the screws and the surgical cut has healed. There are no new fractures. The spaces in your joints are normal and there's no arthritis. However, your bones appear a bit weaker than usual, likely	FINDINGS: This report is about a follow-up examination of your ankle and foot after you've had some procedures done. Starting with your ligaments, which are the tough bands of tissue that connect your bones, the ones at the front and back of your ankle (anterior and posterior tibiofibular ligaments) are intact. Some of your other ligaments have been remodeled, which means they've been surgically altered and have healed. These include the anterior talofibular ligament, calcaneofibular ligament, posterior talofibular ligament, and deep deltoid ligament fibers. Moving on to your tendons, which connect your muscles to your bones, the ones in the front of your ankle are intact. You've had previous repairs to the posterior tibial and flexor digitorum longus tendons, and these are also intact. The tendon that moves your big toe (flexor hallucis longus tendon) and the peroneal tendons on the side of your ankle are intact as well.

(*continued on next page*)

Table 1 (*continued*)

	X-Ray foot	CT ankle	MRI ankle
	foot (first interphalangeal joint) is completely fused together (ankylosis), which was also seen in your previous examination. You have severe arthritis in the joint between your shin bone and your ankle bone (talocrural joint). The surfaces of these bones are touching each other (bone-on-bone apposition), which is a sign of severe arthritis. The bone underneath the cartilage (subchondral bone) is denser than usual (sclerosis), and the surface of the bone is irregular. There are also bone spurs (osteophytes) around the joint. There's a small amount of swelling in the soft tissues around your ankle and the top of your foot. In summary, your foot and ankle are showing signs of healing from the triple arthrodesis surgery, but there's a small area of less dense bone at one of the fused joints and around one of the screws. You have moderate to severe arthritis in several foot and ankle joints, which hasn't changed since your last examination. There's also a small amount of swelling in the soft tissues around your ankle and the top of your foot.	because they haven't been used as much. The CT scan isn't the best tool for looking at the ligaments or tendons in your ankle, but from what we can see, there don't seem to be any major issues with these structures. The tissues just under your skin look normal. In summary, your ankle is healing well after the procedures. The graft has incorporated well into your original bone and there are no complications from the surgical cut. However, your bones are a bit weaker than usual, likely due to not using them as much.	In terms of your bones and joints, you've had a surgical cut (osteotomy) made in your heel bone (calcaneus) that has healed. There's been some bone remodeling at the front of your shin bone (anterior distal tibia) and the bone on the inside of your foot (medial navicular). You have a defect in the bone and cartilage at the inside part of your ankle bone (medial talar dome) that measures 1.1 cm in length and .6 cm in depth. This has been previously repaired with a bone graft. However, the cartilage over this area is completely worn away (full-thickness chondral loss), and there's a small cyst and some mild swelling in the bone underneath (subchondral cyst and edema). There are small bone spurs (osteophytes) in your ankle joint (tibiotalar joint). There's also some swelling in the bone marrow of your ankle bone (posterior talus) and around the sinus tarsi, which is a small canal in your foot. There's a small cyst in your heel bone (calcaneus) in this area. The cartilage in the joint between your heel bone and your ankle bone (posterior facet of the subtalar joint) is thinning, and there are areas of bone remodeling. You have arthritis in several joints in your foot, including the talonavicular and calcaneocuboid joints. Your plantar fascia, which is a band of tissue that runs along the bottom of your foot, is not torn. There's no mass in the tarsal tunnel, which is a space on the inside of your ankle. There are some artifacts, or distortions, in the images due to the screws and metallic staple from your previous surgeries. In summary, your ligaments and tendons are intact after your previous surgeries, and your bones are healing. However, there's a defect in your ankle bone that's been previously repaired, but the cartilage over this area is completely worn away and there's a small cyst and some mild swelling in the bone underneath. There's also some arthritis in several of your foot joints.

AI-LLM, Artificial Intelligence Large Language Model; CT, computed tomography; MRI, magnetic resonance imaging

statistically significant higher mean report length in the CT report compared to the X-Ray report (95% CI: [127.7, 140.0]; $p < 0.001$), in the MRI report compared to the X-Ray report (95% CI: [147.9, 160.2]; $p < 0.001$) and MRI report compared to the CT report (95% CI: [14.03, 26.33] $p < 0.001$).

With regards to the AI-LLM generated reports, one-way ANOVA demonstrated statistically significant differences in FRES scores between 2 report types ($F = 44.9$, $p < 0.001$) (Fig. 4). There was statistically significant higher mean FRES scores in the X-Ray reports compared to the CT reports (95% CI: [3.6, 6.8]; $p < 0.001$), and in the X-Ray reports compared to MRI reports (95% CI: [4.2, 7.4]; $p < 0.001$). There was no statistically significant difference in mean FRES scores between CT reports and MRI reports (95% CI: [−0.9, 2.2] $p = 0.5995$). One-way ANOVA demonstrated no statistically significant difference in mean FKRL scores between all report types ($F = 2.0$, $p = 0.13$) (Fig. 5). One-way ANOVA demonstrated statistically significant differences in mean report length between 2 report types ($F = 806.2$, $p < 0.001$) (Fig. 6). There was a statistically significant higher mean report length in the CT report compared to the X-Ray report (95% CI: [105.4, 120.8]; $p < 0.001$), in the MRI report

Table 2
Mean Flesch Reading Ease Scores, Mean Flesch-Kincaid Reading Level and Mean Length of Report of the Original Radiology Reports and the Artificial Intelligence Large Language Model Generated Report.

	FRES score of original report	FRES score of AI-LLM generated report	Significance	FKRL score of original report	FKRL score of AI-LLM generated report	Significance	Length of original report (words)	Length of AI-LLM generated report (words)	Significance
X-Ray (n = 100)	33.8 ± 6.8	72.7 ± 5.4	p < 0.001	12.2 ± 1.1	8.5 ± 0.4	p < 0.001	37.6 ± 11.1	179.5 ± 33.8	p < 0.001
CT (n = 100)	27.8 ± 4.6	67.5 ± 4.9	p < 0.001	15.4 ± 2.0	8.4 ± 0.6	p < 0.001	171.4 ± 23.5	292.2 ± 16.9	p < 0.001
MRI (n = 100)	20.3 ± 7.2	66.9 ± 3.9	p < 0.001	14.1 ± 1.6	8.5 ± 0.5	p < 0.001	191.6 ± 18.6	293.5 ± 13.5	p < 0.001

CT, computed tomography; MRI, magnetic resonance imaging; FRES, Flesch reading ease score; FKRL, Flesch-Kincaid reading level; AI-LLM, Artificial Intelligence Large Language Model; n, number

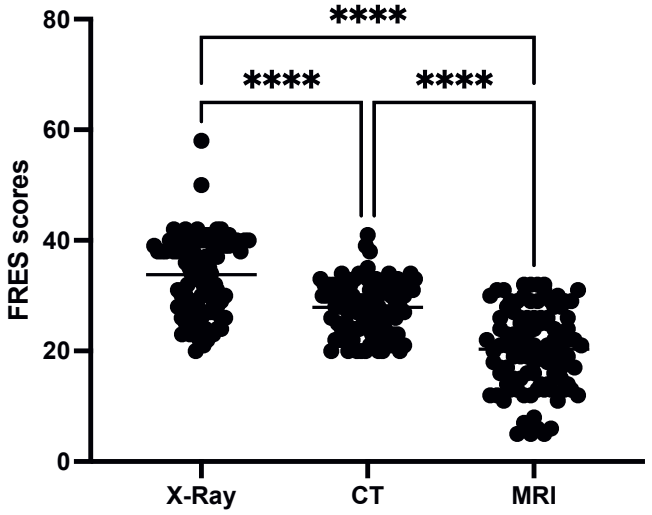


Fig. 1. One-way Analysis of Variance comparing Flesch Readability Ease Scores between the original X-Ray reports, Computed Tomography reports and Magnetic Resonance Imaging reports. **** denotes p < 0.0001.

compared to the X-Ray report (95% CI: [106.5, 121.9]; p < 0.001). There was no statistically significant difference in mean report length in the MRI reports compared to the CT reports (95% CI: [−8.8, 6.6] p = 0.9384).

3.5. Accuracy of AI-LLM report types

Mean Likert Scores with regards to accuracy of the AI-LLM report type is listed in Table 3. The mean Likert score for the X-Ray AI-LLM report was 4.0 ± 0.3, the mean Likert score for the CT AI-LLM report was 3.9 ± 0.4, and the mean Likert score for the MRI AI-LLM report was 3.9 ± 0.4. One-way ANOVA demonstrated statistically significant differences in mean Likert scores between 2 report types (F = 3.9, p = 0.02) (Fig. 7). There was a statistically significant higher Likert score for the X-Ray report compared to the CT report (95% CI: [0.002, 0.2]; p = 0.046), and in the X-Ray report compared to the MRI report (95% CI: [0.01, 0.3]; p = 0.027). There was no statistically

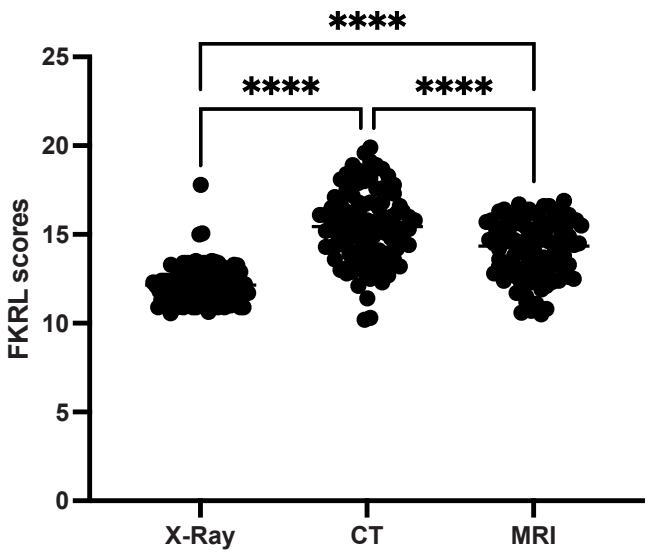


Fig. 2. One-way Analysis of Variance comparing Flesch Kincaid Readability Level scores between the original X-Ray reports, Computed Tomography reports and Magnetic Resonance Imaging reports. **** denotes p < 0.0001.

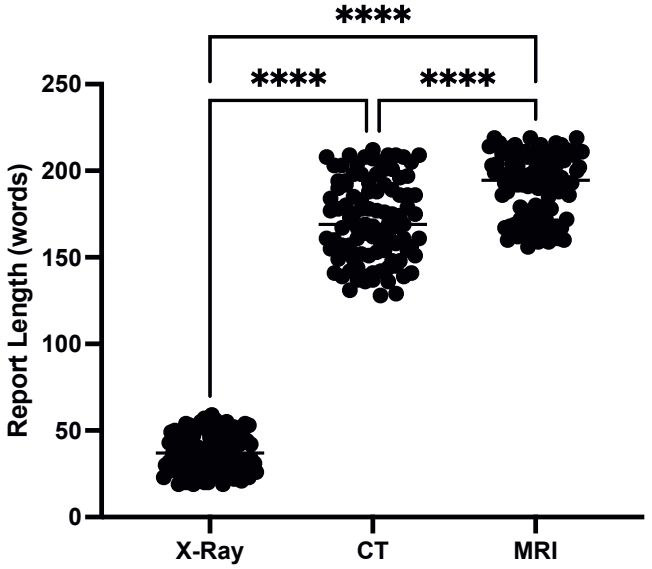


Fig. 3. One-way Analysis of Variance comparing mean report length between the original X-Ray reports, Computed Tomography reports and Magnetic Resonance Imaging reports. **** denotes p < 0.0001.

significant difference in mean Likert scores between the MRI reports compared to the CT reports (95% CI: −0.11, 0.13] p = 0.9783).

Hallucinations were detected in 4 X-Ray reports (4%), 7 CT-scan reports (7.0%) and 6 MRI reports (6.0%) (Table 3). There was no statistically significant difference in the rate of hallucinations between the 3 radiology reports.

4. Discussion

The most important finding of this study was that an AI-LLM was an efficacious tool for improving the readability of reports of foot and ankle radiological examinations across multiple imaging modalities. Superior FRES scores together with superior Likert scores were observed in the X-Ray AI-LLM reports compared to the CT and MRI AI-LLM reports. This study demonstrates the potential use of

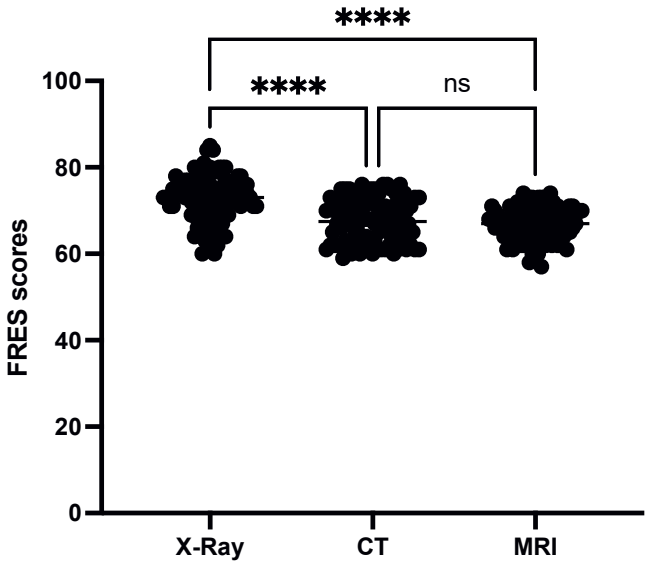


Fig. 4. One-way Analysis of Variance comparing Flesch Readability Ease Scores between the Artificial Intelligence Large Language Model generated X-Ray reports, Computed Tomography reports and Magnetic Resonance Imaging reports. **** denotes p < 0.0001.

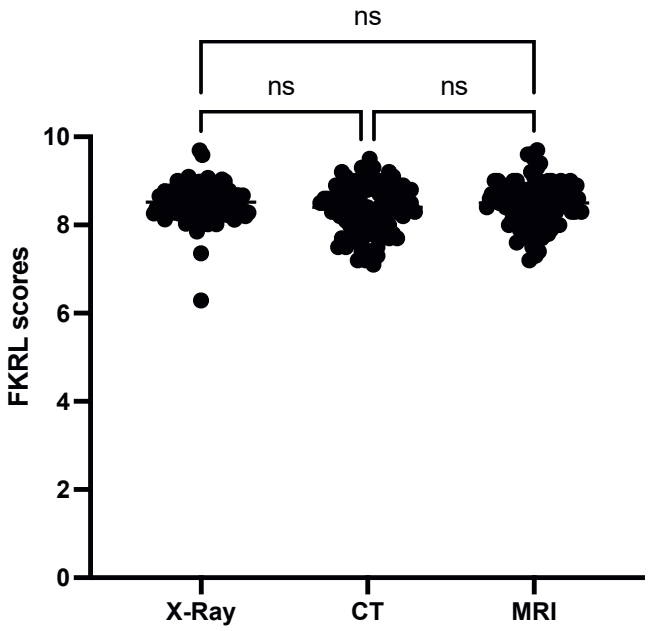


Fig. 5. One-way Analysis of Variance comparing Flesch Kincaid Readability Level scores between the Artificial Intelligence Large Language Model generated X-Ray reports, Computed Tomography reports and Magnetic Resonance Imaging reports. **** denotes $p < 0.0001$. ns denotes not statistically significant.

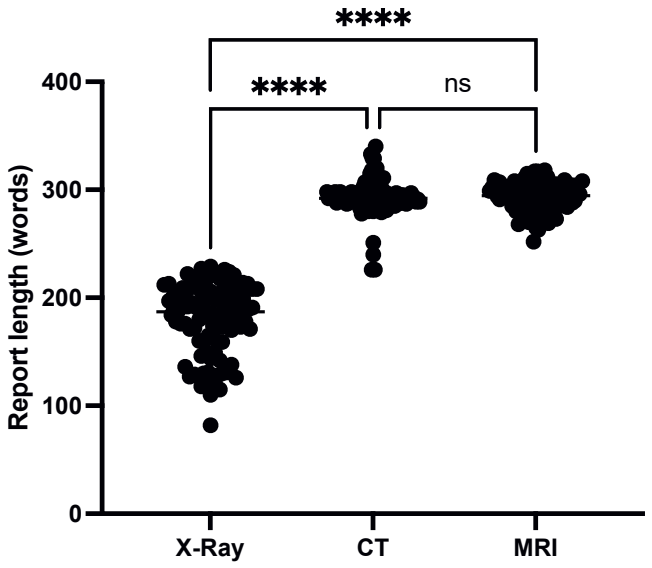


Fig. 6. One-way Analysis of Variance comparing mean report length between the Artificial Intelligence Large Language Model generated X-Ray reports, Computed Tomography reports and Magnetic Resonance Imaging reports. **** denotes $p < 0.0001$. ns denotes not statistically significant.

Table 3
Mean Likert Scores, and number of hallucinations recorded in the Original Radiology Reports and the Artificial Intelligence Large Language Model Generated Report.

	Likert score of AI-LLM generated report	Hallucinations of AI-LLM generated report (N, %)
X-Ray (n = 100)	4.0 ± 0.3	4 (4.0%)
CT (n = 100)	3.9 ± 0.4	7 (7.0%)
MRI (n = 100)	3.9 ± 0.4	6 (6.0%)

AI-LLM, Artificial Intelligence Large Language Model; CT, computed tomography; MRI, magnetic resonance imaging; N, number

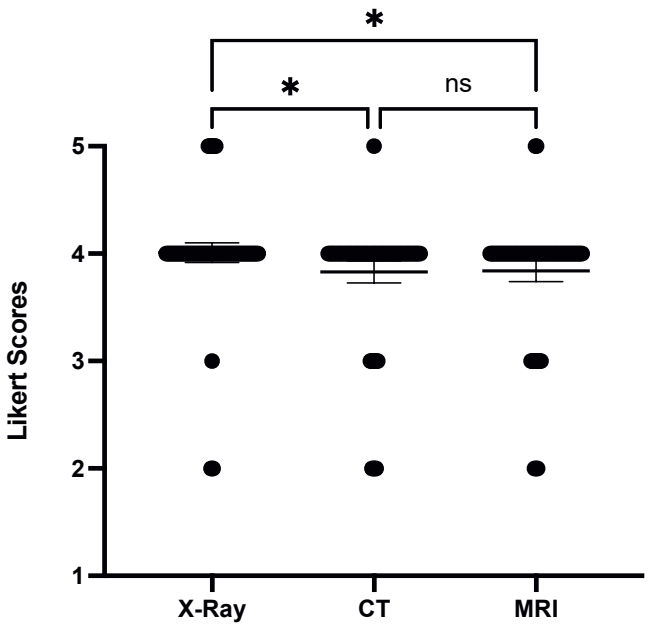


Fig. 7. One-way Analysis of Variance comparing mean Likert Scores between the Artificial Intelligence Large Language Model generated X-Ray reports, Computed Tomography reports and Magnetic Resonance Imaging reports. **** denotes $p < 0.0001$. ns denotes not statistically significant.

AI-LLMs as a new patient-centric approach for enhancing patient understanding of their imaging reports.

Patient anxiety is commonly encountered in the setting of radiological examinations. A cross-sectional study of 488 patients undergoing various diagnostic imaging procedures by Forshaw et al found that approximately 49% of the patient cohort experienced raised anxiety levels directly related to the imaging procedure [7]. Furthermore, the immediate dissemination of radiology reports to patients via the 21st Century Cures Act has further added to the burden of stress and anxiety experienced by patients. Radiology reports are inherently difficult to understand by non-medically trained personnel due to the complex medical terminology and fractured syntax. A study by Martin-Carreras et al found that only 4.2% of radiology reports were readable at or below the mean reading level of the US adult (Grade 8) [8]. Consequentially, Keselman et al reported that only 45% of patients found that their radiology report was “easy” to comprehend [9]. The potential for misinterpreting the findings listed in a radiology report, primarily due to the high readability demands, has been demonstrated to be a deterrent for patients voluntarily accessing their radiology reports [10].

In light of the significant stress and anxiety attributed to these complex radiology reports, various efforts have been made to enhance patient understanding of their imaging reports such as including definition of key terms and embedded diagrammatic pictures. Cook et al. observed that 82% of patients found that the embedded definitions for key terms within their imaging reports were deemed easily comprehensible. Moreover, 77% of participants reported that these definitions enhanced their understanding of the report [2]. Additionally, Recht et al found that the utilization of video radiology reports was well received by patients with 91% of the cohort preferring the use of both written and video radiology reports [11]. However, despite the success of these patient-centered approaches, they are limited by the additional time-intensive burden placed on physicians, thus potentially reducing the mean number of imaging reports that could be potentially generated per unit time.

This current study found that AI-LLM can be utilized as an effective tool for improving the readability of radiology reports for

non-medically trained patients. The mean FRES score and FKRL scores improved significantly across X-Ray reports, CT reports and MRI reports, suggesting a general applicability of AI-LLM in making complex radiological information more accessible to the general populace. For FRES scores, there was a mean improvement of 38.9, 39.8 and 46.6 for the AI-LLM generated X-Ray, CT and MRI reports, respectively. For FKRL scores, there was a mean improvement of 3.7, 7.0 and 5.6, for the AI-LLM generated X-Ray, CT and MRI reports, respectively. One-way ANOVA demonstrated superior mean FRES scores in the AI-LLM generated X-Ray report compared to the CT reports and MRI reports. The original X-Ray reports consisted of fewer words compared to the original CT reports and MRI reports, leaving less medical jargon to be explained in the AI-LLM generated report, thus decreasing the FRES score. Additionally, there appeared to be more X-Ray reports with no pathological findings in comparison to the CT reports and MRI reports, further improving the readability of the report.

Overall, the mean readability for each AI-LLM generated report type was grade 8, which is at the mean reading level for a US adult [12]. Although the National Institute of Health recommends that health information provided to patients should be at a grade 8 or below, the American Medical Association advocates that patient health related materials to be grade 6 or below [12]. We expected the mean reading level to be lower than the mean reading level of grade 8 observed in our study, which can largely be attributed to 2 factors. We chose the following prompt: “Explain this radiology report to a patient in layman’s terms in the second person: < Report Text > ”. This prompt produced reports which explained each individual complex medical term, many of which had a high syllable count thus increasing the FRES score and FKRL score. Furthermore, in an effort to explain each term, this consequentially increased the sentence length, further increasing the FRES score and FKRL score. Although the mean FRES score and FKRL score were higher than initially anticipated, the end of each AI-LLM generated report included a 2–5 sentence summary highlighting and reinforcing the key findings of the report in simple, layman’s terms without any complex medical terminology.

A draw-back associated with AI-LLMs is the potential generation of hallucinations [13]. Hallucinations occur when an AI-LLM produces text that is contextually or factually incorrect, misleading, or unrelated to the input provided [13]. Hallucinations may be the result of the model drawing from its training data and generating content that may seem plausible but lacks accuracy. Overall, the rate of hallucinations recorded in this study was low (range, 4–7%). The hallucinations produced by the AI-LLM in this study could be potentially harmful to the respective patient. Relevant examples include incorrectly stating that a patient had a lateral ligament complex tear when the lateral ligament complex was intact, or claiming that there was no fracture of the lateral malleolus when a fracture was present. As such, patients should not rely solely on the use of AI-LLMs for the interpretation of their radiology reports, but rather should always have a detailed discussion of the clinical relevance of both the report and images with the ordering physician.

This study has numerous biases and limitations. Firstly, only 1 AI-LLM version was utilized thus, the findings of this study may not be applicable to different versions and configurations of various AI-LLMs. Secondly, the readability of the report was determined only by historical readability indices. Further studies assessing the readability of the AI-LLM generated reports by patients should be conducted to obtain a more robust insight into the readability of these reports. Finally, these reports were obtained only from the foot and ankle, and, therefore, cannot be generalizable to radiological reports from other joints.

5. Conclusion

The most important finding of this study was that AI-LLM was an efficacious tool for improving the readability of reports of foot and ankle radiological examinations across multiple imaging modalities. Superior FRES scores together with superior Likert scores were observed in the X-Ray AI-LLM reports compared to the CT and MRI AI-LLM reports. This study demonstrates the potential use of AI-LLMs as a new patient-centric approach for enhancing patient understanding of their imaging reports.

Declaration of Competing Interest

John G. Kennedy is a consultant for Arteriocyte, In2Bones and Arthrex. John G. Kennedy receives financial support from the Ohnell Family Foundation, Mr Winston Fischer and Ms Tatiana Rybak.

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